

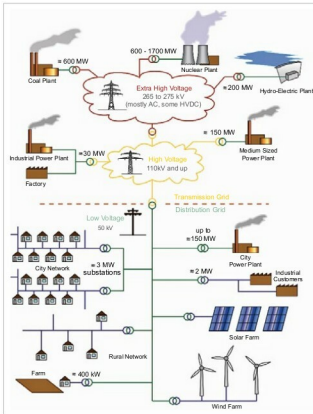


Figure 1: Typical power system network

After downloading, you can check the downloaded file in your home directory. To convert it to a Deb file, you need to install the alien package, as follows:

```
hithu@linux:$ sudo apt-get install alien
```

Convert the rpm to Deb by using the command given below:

```
hithu@linux:$ alien gridlabd-3.2.0-1.x8664.rpm
```

Exit the terminal. Right-click and open the newly generated Debian file using the Ubuntu software center package manager to complete the installation.

Check the installation by using the following command:

```
hithu@linux:$ gridlabd --version
GridLAB-D 3.2.0-5368 (Jojoba) 64-bit LINUX RELEASE
```

To try a simulation of residential temperature variations for a year, save the code given below as *residential.glm* using a text editor (gedit):

```
clock {
    starttime '2017-02-01 00:00:00 UTC';
    stoptime '2018-02-01 00:00:00 UTC';
```

```
}
module residential;
module tape;
object house {
    object recorder {
        property air_temperature;
        file temperature.csv;
    };
};
```

In terminal, run the file:

```
asgridlabd residential.glm
```

To see output:

```
more temperature.csv
2017-02-01 00:00:00 UTC,+72.4767
2017-02-01 01:00:00 UTC,+73.5292
2017-02-01 02:00:00 UTC,+74.1789
2017-02-01 03:00:00 UTC,+74.7188
2017-02-01 04:00:00 UTC,+75.1863
2017-02-01 05:00:00 UTC,+75.6680
2017-02-01 05:24:12 UTC,+76.0000
2017-02-01 05:29:10 UTC,+73.9968
2017-02-01 06:00:00 UTC,+75.9057
2017-02-01 06:02:18 UTC,+76.0006
2017-02-01 06:07:40 UTC,+73.9994
2017-02-01 06:25:12 UTC,+76.0006
2017-02-01 06:30:33 UTC,+73.9961
2017-02-01 06:48:21 UTC,+76.0001...
```

Hence, you have the result of temperature variations in a typical house with hourly intervals, for a year. It is possible to change the interval, add the location of meteorological data, measure various parameters, add more implicit devices, and much more.

There are many modules available within this software:

```
gridlabd --modhelp residential
```

The module given above the list of the various classes and objects possible in a residential simulation. For example, class 'freezer' has...

```
class freezer {
    parent residential_enduse;
    class residential_enduse {
        loadshape shape;
        enduse load; // the enduse load description
        complex energy[kVAh];
    } // the total energy consumed since the last meter reading
    complex power[kVA];
    // the total power consumption of the load
```